LONGEVITY LEVEL 3

LEVEL 1

SMOKING

Full Leaderboard Data

Mouse Lifespan Leaderboard

Hover over an intervention name for more details.

#		INTERVENTION	HUMAN AGE EQUIV-ALENT START	MAX LIFESPAN INCREASE	AVG LIFESPAN INCREASE
1	CR	Caloric restriction 25% human equivalent	45	54%	62%
2	TLT	6 x Telomerase Gene Therapy IV (CMV) Retrot Study	55	41%	41%
3	PDA	20 x PDA-001 Placental Stem Cells IV	30	32%	21%
4	FGF	6 x Follistatin Gene Therapy IV	55	29%	33%
5	NAC	NAC + Glycine 1.2g each	46	25%	24%
6	RA	9 mg Rapamycin/wk + up to 600 mg acarbose/d	28	24%	37%

ORAL HEALTH OVERVIEW

- **INVISALIGN EQUIVALENT £700**
  www.alignerco.com
- **PRO TEETH WHITENING FOR £100**
  NITEWITE www.zoomscotland.com, including custom retainers
- **DENTAL HYGIENIST**
  Scale/root plane, Air polishing is great, advanced stain removal
- **BONDING/VENEERS**
- **REMOVE ALL MERCURY SAFELY**
  --> ceramic/gold
- **FIX REFLUX & SIBO/IMO**
  IMT PowerBreath, Reflux Food-Drink Avoidance List, PPI trial
- **TOOTHBRUSH**
  BURST Pro (with world-class tongue scraper)
- **SMARTFLOSS**
  Expanding microplastic free floss
- **WATERPIK WITH H2O2**
  Amazon your choice
- **BIOMIN F (NANOFLUORAPETITE)**
  www.biomin.co.uk ; or Biomin C (Sulphate & F free)
- **XRAYS - OVERUSED**
  Vitamin C 1g to reduce DNA damage (potentially)
  Lead free neck and head collars for thyroid/brain cancer

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LEVEL 1

PURE AIR

INTRODUCTORY CONCEPTS

LEVEL 1

DIET COMPOSITION

PLEASE SCAN

oliverzolman.com/calculators

500+ ZOLMAN 'CLOCKS'

E.G.

VO2 MAX = 1 CLOCK

GRIP STRENGTH = 1 CLOCK

HEART ULTRASOUND AGE (20 MARKERS COMBINED) = 1 CLOCK

PWV = 1 CLOCK

VISIA FACE SCAN = 1 CLOCK

Phenotypic clocks ≠ epigenetic

EXAMPLES

MAX 2.5 HRS VIGOROUS A WEEK IS SAFE (?)

What counts for the 6+ hours exercise target per week? Only moderate & vigorous.

MODERATE EXAMPLES (~HR 110-150)

- Brisk walking - Above 6 kmph (2 steps per second)
- Housework - Washing windows, vacuuming, mopping, lawnmowing, DIY
- Cycling light/moderately - 8 - 16 kmph (5 - 10 mph)
- Badminton singles
- Tennis doubles
- Rowing at 50 - 75 watts
- Strength training

VIGOROUS EXAMPLES (~HR 150+)

- Jogging or hiking - Jogging over 8 kmph (5 mph)
- Carrying heavy loads
- Tennis singles
- Football
- Basketball
- Rowing at > 75 Watts
- Cycling fast / or vigorously (> 10 mph)
- Bootcamps / Circuits

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MODERATE
(3 - 6 METs)

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VIGOROUS
(> 6 METs)

ZOLMAN LONGEVITY LEVEL 2: OPTIMISE ALL NON-AGE-RELATED FACTORS

1. **YOUR HEALTH ROLODEX**
   World's Top Professionals | Friends & colleagues | Family
2. **OPTIMAL NON-AGEING CONDITION DIAGNOSIS & TREATMENT**
   Auto-immune | Rare diseases | Chronic Pain | More
3. **SEX & SEXUAL HEALTH**
   Theory | Vaccines | Drugs & Devices
4. **RECREATIONAL DRUGS PROTOCOLS**
   Alcohol | Stimulants | Downers
5. **AVOIDING HARM**
   Clinicians | Therapies | People | Natural Disasters | Rabid dogs
6. **EVIDENCE BASED MEDICINE (EBM) SKILLS**
   Searching | Reading | Making decisions
7. **GENOMICS**
   Tests | Polygenic risk | Monogenic results | Drug responses
8. **COGNITIVE ENHANCEMENT**
   Hardware | Software | Prescriptions/Drugs | Lifestyle
9. **TESTING EQUIPMENT OWNED**
   Devices | Imaging systems | Software & tracking results
10. **ENVIRONMENTAL EXPOSURES**
    Air | Light | Water | Animals | People | Occupation
11. **EMERGENCY MEDICINE**
    Skills | Knowledge | Equipment Owned | Cryopreservation
12. **BIOBANKING**
    Organelles | Cells | Tissues | Organs | Whole body clones

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BASIC SEX HORMONES

This is an oversimplification

MEN

Free T: 0.4 - 0.6 nmol/L (12-17 ng/dL)

Bioavailable T

Gel pump, e.g. Tostran, Testogel

Low dose HCG subcut injections 1-3x/week to preserve testicular T production (mimics exogenous LH)

Higher dose HCG +/- clomiphene (boosts FSH+LH) to preserve sperm production (but ED risk)

Or just sperm bank

Get Tested: Saliva Free T - no blood required, £40

WOMEN

Start body/bioidentical HRT up to 10 years after menopause

= Micronised Progesterone (oral)

= 17Beta-Estradiol (patch, gel, spray, not oral)

Egg bank / ovarian tissue cryopreservation

Can reimplant tissue to delay menopause to age 60 *in theory* - multilple laparoscopies under GA

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LEVEL 1

CRON

Caloric Restriction Optimal Nutrition

INTRODUCTORY CONCEPTS

LEVEL 1

KEY VITAMINS

JAWLINE – AT REST

0	1	2	3	4
No sagging	Mild sagging	Moderate sagging	Severe sagging	Very severe sagging

GOALS OF THIS PRESENTATION

1. UNDERSTAND THE MOST EVIDENCED BASED WAYS TO REACH AGE 80 (LONGEVITY LEVEL 1)
2. UNDERSTAND THE POTENTIAL THERAPIES NEEDED TO GO BEYOND THIS (LONGEVITY LEVEL 3)

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LEVEL 1

HORMONES

9:41
Blood Pressure
Day
Week
Month
08:00 - 10:00
Average of 3 measurements
119 Sys
128-140
77 Dia
128-140
75 bpm
HR
hilo

LEVEL 1

BLOOD PRESSURE

HOW TO MEASURE AGING IN ORGANS

- RESEARCH GRADE MRI
- RESEARCHR GRADE ULTRASOUND
- SKIN IMAGING
- CT SCAN OF HEART ARTERIES
- MEDICAL DEVICES (E.G. LUNG MACHINES, VO2)
- BIOFLUIDS (E.G. BNP, CYSTATIN-C, PHENOAGE)

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LEVEL 1

BMI (AND BODY FAT)

LEVEL 1

EXERCISE

PROJECTS TO ACHIEVE REJUVENATION BY 2030

2ND HAND SMOKE (INNOCENT)

- 1 WORLD TRADE CENTRE A DAY (!)
- 5 INNOCENT FAMILIES KILLED SINCE THIS TALK STARTED

3RD HAND SMOKE

- IF HOTEL PERMITS SMOKING IN OTHER ROOMS, 4X HIGHER CARCINOGEN LEVELS
- SMOKERS HOUSE DUST: 700X HIGHER CARCINOGENS

QUESTIONS

5 minutes

Fig. 9. Twins (natural age 71) with difference in hormone replacement therapy. Twin B (*right*) had 22 more years of hormone replacement therapy than twin A (*left*). Twin B had a 1.2 lower body mass index. Perceived age difference was 7.25

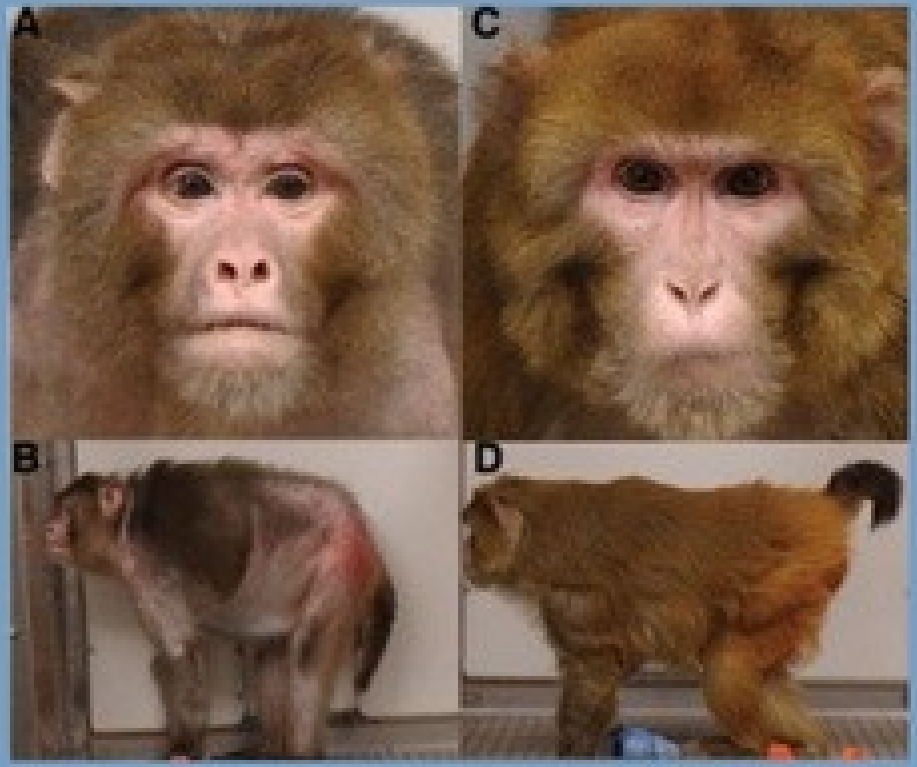
22 years of HRT from age 49 in identical twin ,
same BMI, = 7 years younger

JOIN THE EVIDENCE BASED LONGEVITY WHATSAPP GROUPS

Zolman

WhatsApp contact

- **350 CLINICIANS**
  Send your LinkedIn or other evidence of clinician status to join
- **500 ENTHUSIASTS**


A
C
B
D

SLEEP - QIAN 2023

1. IDEAL SLEEP DURATION OF 7-8/HR/NIGHT
2. DIFFICULTY FALLING ASLEEP NO MORE THAN 2X/WEEK
3. TROUBLE STAYING ASLEEP NO MORE THAN 2X/WEEK
4. NOT USING ANY SLEEP MEDICATION
5. FEELING WELL RESTED AFTER WAKING UP AT LEAST 5 DAYS/WEEK

SCORE 5/5,

LIFE EXPECTANCY +4.7 YRS MEN,

+2.4 YEARS WOMEN

VS 0/5 OR 1/5 SCORE

Qian 2023

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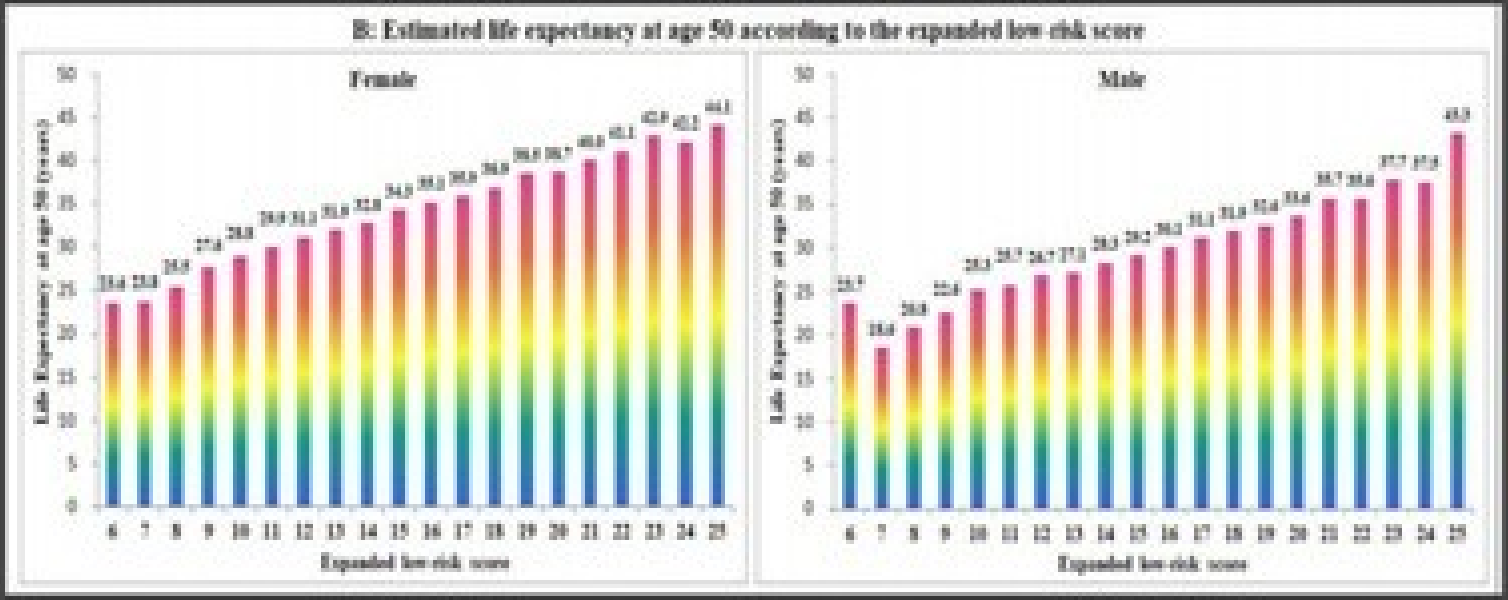
Your body has ran out of magnesium.
Omg

EXAMPLE STUDY WHY LEVEL 1 COMPONENTS STACKS: YANPING 2018 (USA) - 34 YEARS FOLLOWUP (REF)



RETROSPECTIVE EPIDEMIOLOGY

Research Article | Open Access
2021 | Volume 2 | Issue 6 | https://doi.org/10.37191/Mapsci-2582-3930-2(6)-047

Safety Study of AAV hTERT and KLOTHO gene transfer therapy for dementia

Received	Revised	Accepted	Published
Sep 20, 2021	Oct 19, 2021	Oct 20, 2021	Nov 20, 2021

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Abstract

A sponsored, interventional, non-randomized study without a control group using a novel and proprietary central nervous system gene transfer method to deliver AAV hTert and Klotho genes to five patients with mild or moderate dementia was performed to primarily evaluate safety. Clinical response data was gathered as a secondary interest. The therapy demonstrated a very high safety profile with no serious adverse effects identified. Clinical evaluation of the patients over the course of the one year follow up yielded significant findings with all five patients demonstrating evident reversal of Dementia symptoms such as sustained cognitive improvement as measured by the Folstein exam. Telomere analysis was performed before and after the therapy. A measurable elongation of the participants telomeres was identified, and biological age was reduced as chronological age increased.

1. ONCE YOU CAN MEASURE YOUR 81 ORGAN AGES:

Measure biological age accurately in each of the 81 organ types in individuals:

Accuracy is enhanced via peer-reviewed Zolman Biological Age Marker (Z-BAM) Criteria

2. YOU CAN SEE IF YOU CAN MAKE THEM YOUNGER (REJUVENATION):

Use Level 3 Therapy Experimental Guidelines: to reverse age of each organ in individuals (ideally AGREE-II and IPDAS compliant)

Realtime Monitoring

Compare with outdoor

Breathe Easy

Track changes over time and see how your actions impact the health of your space.

Personalized Tips and Insight

EXTREMELY IMPORTANT THAT WAIST IS LESS THAN HIPS

ZOLMAN LONGEVITY LEVEL 1:
STACK UP TO 30 YEARS LONGEVITY (?)

- SMOKING: 0 VS 20/DAY +10 YRS
- MENTAL HEALTH: NONE VS SEVERE +10 YRS
- EXERCISE: +6 HRS MOD/VIG A WEEK + 6K STEPS +8 YRS
- CALORIES (CRON): 10-20% +RDA +5 YRS IN MEN ONLY
- AHEI-2010 DIET: SEE THE CHECKLIST +5 YRS
- BMI (BODY FAT MOSTLY) & WAIST HIP RATIO: 18.5 TO 22.5, WAIST LESS THAN HIPS +5 YRS
- BLOOD PRESSURE: UNDER 115/70 VS 140/80 +3 YRS
- VO2 MAX: 50 ML VS 30 ML +2.5 YRS
- HORMONES (T/E/P): FREE T 0.4-0.6 NMOL/L (715-770 USA) (MEN), HRT POST-MENO (FEM) +2 TO 5 YRS
- GUIDELINE RECOMMENDED SCREENING + FIXING: OLIVERZOLMAN.COM/CALCULATORS +2 YRS
- SLEEP: MEET 5 CRITERIA +2 YRS
- VITAMINS: D 75-125 NMOL/L (30-50 NG/ML) + MIDDLE REF RANGE B12 & FOLATE: +2 YRS
- SOCIAL STRENGTH: STRONG BONDS & PURPOSE VS ISOLATED & LOST +2 YRS
- PERFECT AIR QUALITY + PERFECT ORAL HEALTH: + 1 YEAR EACH
- ALCOHOL: 7 DRINKS/WEEK TO 1 +7 MONTHS (UNLESS HIGH RISK, E.G. CANCER FAMILY HISTORY)

DO THEM ALL

= AGE 90 (?)

TO GET BEYOND 90 (?):

PHARMACEUTICALS, GENE THEARAPY, CELL THERAPY, NANOTECH, LAB GROWN ORGANS, WORKING CRYO.

+ avoid AI p(doom)

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3 BASIC VITAMIN TESTS = LESS WREKD

Randox UK units

B12: 500+ PG/ML

= 370+ pmol/L

FOLATE: 7+ UG/L

Methylfolate supplements or food only

= 15+ nmol/L

D: 75 - 125 NMOL/L

= 30 - 50 ng/mL

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EVIDENCE BASED LONGEVITY THE LONGEVITY LEVEL 1, 2, 3 PROTOCOL

Dr Oliver N. G. Y. Zolman MBBS BSc - Medical Doctor

This presentation has been peer-reviewed

Our Gene Therapy Arsenal

Klotho

Master aging regulator - improves kidney function, cognitive performance, calcium metabolism

Follistatin

Muscle growth & metabolism - counteracts myostatin, prevents muscle wasting

hTERT

Telomerase activation - extends cellular lifespan, reverses aging markers

PGC1-α

Mitochondrial biogenesis - enhances energy production, metabolic health

VEGF

Vascular regeneration - improves blood flow, tissue repair, wound healing

FGF21

Metabolic regulation - glucose control, fat metabolism, liver protection

SIRT1

Cellular longevity pathways - DNA repair, inflammation control

FOXO3

Stress resistance & lifespan - cellular protection, tumor suppression

P53

Tumor suppressor restoration - prevents cancer, induces cell repair

SIRT6

DNA repair & inflammation control - enhances genomic stability, extends lifespan

NAMPT

NAD+ enhancement - boosts cellular energy, activates longevity pathways

Case Report | Open Access
2022 | Volume 4 | Issue 2 | https://doi.org/10.37191/Mapsci-2582-3859-4(2)-106

Systemic Human hTERT AAV Gene Transfer Therapy and the Effect on Telomere Length and Biological Age, A Case Report

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[2]Biovivia, USA

***Corresponding Author**: Patrick E Sewell, Integrated Health Systems, Biovivia, USA.

Received	Revised	Accepted	Published
Mar 19, 2022	Mar 26, 2022	Apr 5, 2022	Apr 29, 2022

Abstract

A single adult female human was treated with AAV hTERT gene transfer therapy on two separate occasions 5 years apart. Follow-up is 5.8 years. The first dose administered on 9.16.2015 consisted of a total intravenous dose of 3e15 AAV hTERT. The second dose administered on 9.24.2020 consisted of a total intravenous dose of 3e15 AAV hTERT. Before and after each therapy and periodically in between the doses, Human Leucocyte telomere analysis was performed. The initial telomere length measurements collected on 9.15.2015 at the initiation of the AAV hTERT gene transfer therapy demonstrated a baseline average telomere length of 6.71kb which corresponded to a telomere percentage relative to age and population at the 30th percentile. The most recent telomere length measurements collected on 7.13.2021 demonstrated an average telomere length of 8.94kb which corresponded to a telomere percentage relative to age and population at the 89th percentile. The results demonstrate the progressive lengthening of the recipient's telomeres from 6.71kb to 8.94kb despite advancing 5.8 years in chronological age. Associated age as related to telomere length (also known as biological age) was calculated and compared to chronological age. Initially on 9.15.2015, and prior to the first AAV hTERT gene transfer therapy, the associated age was calculated to be 62 years. The latest telomere analysis dated 7.13.2021 demonstrated a calculated associated age of 35 years. This decrease in associated, or biological age, decreased at a rate of 5.3 years per year of chronological age advancement.

PLEASE USE A STRAP-ON-HRM

1. THE WRIST AND FINGER DEVICES CAN BE SO INACCURATE YOU'LL MISS MOD/VIG
2. USE THE CHEST-STRAP-ON-HEART-RATE MONITOR FOR A MONTH TO VALIDATE ACCURACY OF ANY WRIST OR FINGER DEVICE - OR GYM ECG HANDLES
3. BICEP WORN DEVICES MAY BE MORE ACCURATE THAN WRIST AND FINGER, MAYBE AS ACCURATE AS CHEST

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DIET COMPOSITION. +5 YEARS?

CAN CHEAT ON 2 PROBABLY

BLUE GOOD

		Ideal per day	Good amount per day	Tips
1	Vegetables (but not white potatoes)	> 600g veg/1200g leafy veg	450g veg/900g leafy veg	Weigh food. Frozen/microwave veg etc.
2	Fruits	4+ medium fruits/500g berries	3 medium fruits/400g berries	Weigh so can eyeball it in future
3	Wholegrains or legumes	> 90g	> 60g	Chickpeans, lentils, beans.
4	Nuts	> 30g nuts/15g nut butter	20- 30g whole/10-15 g butter	Weigh nuts so can eyeball it in future
5	Omega 6	15 - 20g	15 - 20g	Check packets. Nuts, seeds, oils often.
6	Omega 3	> 250 mg EPA/DHA or 2g ALA	200 mg EPA/DHA or 1.5g ALA	2T ground flaxseed per day, or 100g wild salmon per week
7	Added sugar drinks OR unfresh fruit juice	Never	50 ml per day or 1 glass per week	Small 100 ml, freshly squeezed juice with fibre, no plastic, may be OK 3x/wk
8	All red meat, or white processed meat	Never	5g/d or 30g/week	White meat instead
9	Trans fat	< 1g (dairy OK)	< 2 g (dairy OK)	Check packets
10	Sodium "Salt" NaCl sodium equivalent	< 1.6g men < 1.1g women < 4g men, < 2.8g women	< 2.4g men < 1.6g women < 6g men, < 4g women	Check packets

YELLOW BAD

PURE CARNIVORE? CAN MAKE IT WORK

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WHAT EVIDENCE AND GUIDELINES ARE LONGEVITY LEVEL 1,2,3 BASED ON?

2018

FIRST PUBLISHED

1 **NEW GUIDELINES**

In-house systematic review, peer-review and guideline development process based off of Cochrane, NICE, AGREE-II and IPDAS.

2 **EXISTING GUIDELINES**

Where existing high-quality guidelines exist, these are integrated.

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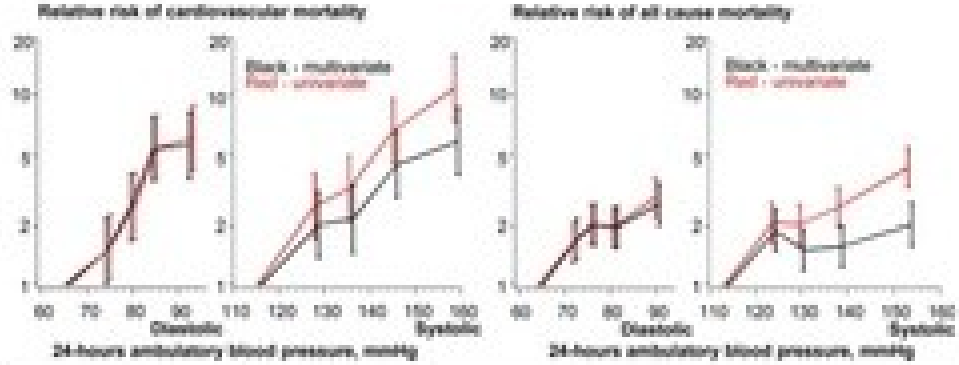
BLOOD PRESSURE + 3 YEARS?



LESS GAINS IF DON'T HAVE EXISTING ATHEROSCLEROSIS

2018 meta analysis of Randomised Controlled Trials ( )

- **SPRINT 10K PERSON RCT (STUDY)**
  - 0.5 - 3 years gain in lifespan for home resting systolic blood pressure mean of 3 readings being under 120 vs 140
  - Depending if you start lowering BP at age 65 or age 50 onwards
- **TARGET 110/70**

  Home resting reading

  take 3 readings

  discard first

  take avg of next two

  ideally as a 24 hour reading with HiLo bracelet

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LEVEL 3 THERAPIES EXAMPLES:

- **LEVEL 3.1: ON-LABEL (LEGAL SOMEWHERE IN THE WORLD)**

  LASER skin rejuvenation, botox, LASER joint rejuvenation, PRP/HA/BM-MSC joint rejuvenation, statins, anti-amyloid Alzheimer's injections, heart transplant, acarbose, inclisiran, evolucumab, tirzepatide/mounjaro, semaglutide/ozempic, statins, pneumococcal vaccine, influenza vaccine, brain shockwave

- **LEVEL 3.2: OFF-LABEL (INDIRECT USE)**

  Rapamycin, metformin in non-(pre)diabetics, GLPs in lower BMIs

- **LEVEL 3.3: UNLICENSED (PRE-PHASE 3 TRIALS)**

  Pelacarsen, BPC-157, TripleHelix gene therapy, brain ultrasound, cryonics

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LONGEVITY LEVEL 1

15 Components

BODY MASS INDEX (BMI)

WEIGHT IN KG / (HEIGHT IN M)^2

BODY FAT %

USING BIA SCALES (BIOIMPEDANCE)

18.5 - 22.5

ITS RARE TO SEE SOMEONE WITH OPTIMAL MARKERS EVEN IF PURE MUSCLE ABOVE BMI 22.5

Non-caucasian ref range is already 18.5 - 22.5

15% MAX MEN
27% MAX WOMEN

Women Body Fat Percentages
10-12%
15-17%
20-22%
25%
30%
35%
40%
45%
50%

Men Body Fat Percentages
3-4%
6-7%
10-12%
15%
20%
25%
30%
35%
40%

BEEPER APP

And News Feed Eradicator
Chrome Extension

MENTAL HEALTH BASICS

Psych 1hr vid consults: £250-370

happyclinic.co.uk - £375

Professor El-Missiry - £250

TopDoctors.co.uk

GP MUCH CHEAPER

Not RE more complex conditions such as Schizophrenia, Cyclothymia, Bipolar etc.

- **ALL OTHER LEVEL 1**
  D, B12, folate, exercise, diet, sleep, hormones... also iron
- **PROMETHAZINE / CLONAZEPAM 0.25-0.5 MG**
  Benzos, max 4-7 days a month, not long term or higher dose ideally
- **SSRI - ESCITALOPRAM**
  Can start 1 mg, or 2.5 mg, build up to 5 mg - 20 mg, can split dose
- **PROPANOLOL**
- **MIRTAZAPINE LOW VS HIGH DOSE**
  For sleep at low dose vs stimulant at high dose
- **10K LUX + 630-660 / 810-850 NM LIGHT MW CHECKED**
- **KNOW: Y-BOCS, GAD7, PHQ9, AQ50**
- **GABA, ASHWA, LAVENDER SILEXAN, MELATONIN, CAFFEINE, THEACRINE, ALCAR 2-4G**
- **SAMPHIRENEURO.COM, NEUROMYST.COM, FLOWNEUROSCIENCE.COM, BRIGHT BRAIN INSTITUTE**
- **BUPROPRION OR BUSPIRONE**
  Bupropion also for smoking cessation
- **TMS - 20-30 SESSIONS RTMS/TBS**
  Magstim Horizon 3 (no Helmets). Mostly low mood + OCD, but GAD (anxiety) likely
- **ECT (ELECTRO CONVULSIVE THERAPY) (GA)**
- **OXFORD KETAMINE CLINIC**
  For depression, not anxiety
- **KNOW: DBT, IFS, CBT, ERP, CBT-I, MINDFULNESS, MEDITATION**
- **ADHD DIAGNOSIS - £300**
  www.adhd-360.com
- **LONG COVID -> HBOT 2ATM 20 SESSIONS**
- **SOCIAL FACTORS + BONDS + TOUCH + HRS/DAY**
- **PSILOCYBIN / MDMA PSYCHOTHERAPY (RESEARCH ONLY E.G. IMPERIAL COLLAGE LONDON TRIALS)**
  Only with approved expert clinicians/researchers and followup

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Spots (92%)
Wrinkles (99%)
Texture (96%)
your actual age
30
your TruSkin Age
24
UV Spots (32%)
Brown Spots (77%)
Red Areas (58%)

RESEARCH GOAL BY 2030

Rejuvenate one hundred 75-85 year olds from biologically age 80 to age 60 in all 81 organ types

THE 8 WAYS YOU DIE

81
ORGAN TYPES

1. **TOO FEW CELLS**
   Stem cell exhaustion, cell loss, less mitochondria
2. **TOO MANY CELLS**
   Cancer, zombie cells, fat cells, etc
3. **DAMAGED CELLS**
   Mitochondrial mutations, genomic instability, telomere shortening
4. **WASTE IN CELLS**
   Lipofuscin, disabled macroautophagy
5. **WASTE OUTSIDE CELLS**
   Amyloid build up
6. **DAMAGED SCAFFOLD BETWEEN CELLS**
   Stiffening, AGEs accumulation
7. **GENE EXPRESSION CHANGES**
   Epigenetic drift + noise
8. **BAD COMMS BETWEEN CELLS**
   Dysbiosis, chronic inflammation, deregulated nutrient sensing

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LEVEL 1

VO2 MAX

INTRODUCTORY CONCEPTS

ZOLMAN LONGEVITY SCHOOL LEVEL 2: OPTIMISE ALL NON-AGE RELATED FACTORS

1. **YOUR HEALTH ROLODEX**
   World's Top Professionals | Friends & colleagues | Family
2. **OPTIMAL NON-AGEING CONDITION DIAGNOSIS & TREATMENT**
   Auto-immune | Rare diseases | Chronic Pain | More
3. **SEX & SEXUAL HEALTH**
   Theory | Vaccines | Drugs & Devices
4. **REC. DRUGS SAFETY PROTOCOLS**
   Alcohol | Stimulants | Downers
5. **AVOIDING HARM**
   Clinicians | Therapies | People | Natural Disasters | Rabid dogs
6. **EVIDENCE BASED MEDICINE (EBM) SKILLS**
   Searching | Reading | Making decisions
7. **GENOMICS**
   Tests | Polygenic risk | Monogenic results | Drug responses
8. **COGNITIVE ENHANCEMENT**
   Hardware | Software | Prescriptions/Drugs | Lifestyle
9. **TESTING EQUIPMENT OWNED**
   Devices | Imaging systems | Software & tracking results
10. **ENVIRONMENTAL EXPOSURES**
    Air | Light | Water | Animals | People | Occupation
11. **EMERGENCY MEDICINE**
    Skills | Knowledge | Equipment Owned | Cryopreservation
12. **BIOBANKING**
    Organelles | Cells | Tissues | Organs | Whole body clones

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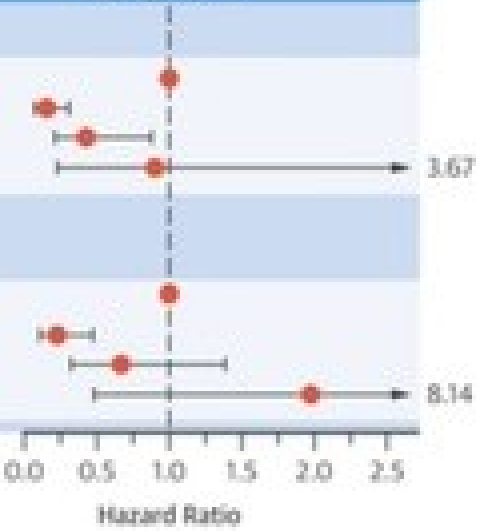
IS MORE THAN 2.5 HOURS VIGOROUS EXERCISE SAFE?

DOSE OF JOGGING	NO. OF PARTICIPANTS	ALL-CAUSE MORTALITY DEATHS	FOREST PLOT
Adjusted for age and sex			
Sedentary nonjogger (reference)	413	128	
Light jogger	576	7	
Moderate jogger	262	8	
Strenuous jogger	40	2	3.67
Adjusted for age, sex, smoking, alcohol intake, education, and diabetes			
Sedentary nonjogger (reference)	394	120	
Light jogger	570	7	
Moderate jogger	252	8	
Strenuous jogger	36	2	8.14



N=5000 MEN & WOMEN

: LIGHT JOGGERS HAD A SLOW OR AVERAGE PACE, APPROXIMATELY 5 MPH (CORRESPONDING TO 6 METS), AND <2.5 H OF JOGGING PER WEEK WITH A FREQUENCY OF 0.5 -3 TIMES PER WEEK.

Zolman 81 Organ Biological Ages

Nervous System
1. Cerebrum
2. Perfusion system: Ventricles, CSF, meninges, subarachnoid space, dural venous sinuses, cerebral veins, Circle of Willis
3. Interhemispheric, Diencephalon (third, lateral, sub-thalamic), basal ganglia, sub-nigra, hippocampus & pineal
4. Pituitary gland
5. Cerebellum
6. Brain stem
7. Spinal cord and central canal
8. Cranial nerves non-Vagus
9. Motor PNS
10. Sensory PNS
11. Autonomic NS

Cardiovascular system
12. Heart
13. Arteries
14. Capillaries
15. Veins

Respiratory system
16. Trachea
17. Larynx
18. Bronchi & bronchioles
19. Lungs
20. Diaphragm

Musculoskeletal system
21. Muscles
22. Visceral fat
23. Subcutaneous
24. Brown fat
25. Beige fat
26. Tendons
27. Ligaments
28. Cartilage & capsule
29. Structural bone (non-marrow)
30. Intervertebral discs

Immune & RBC System
31. Lymph vessels & nodes
32. Red bone marrow
33. Yellow bone marrow
34. Spleen
35. Thymus

Urinary system
36. Kidneys
37. Ureters
38. Bladder
39. Urethra

Ears & eyes systems
40. Inner ears
41. Middle ears
42. Outer ears
43. Retina
44. Lenses
45. The rest of the eye
46. Lids, brows & lashes

Hormonal (both sexes)
47. Adrenal glands
48. Thyroid
49. Parathyroid

Reproductive male system
50. Prostate
51. Seminal vesicles
52. Testicles
53. Epididymides
54. Vas deferens
55. Scrotum
56. Penis

Oral Gastrointestinal Tract
57. Nose
58. Inner ear/sinus
59. Lips
60. Tongue
61. Teeth & dental bone
62. Gums
63. Salivary glands
64. Pharynx
65. Oesophagus
66. Stomach
67. Small bowel
68. Large bowel
69. Mesentery
70. Rectum & anal canal
71. Pancreas
72. Liver
73. Gall Bladder

Skin, hair, nails system
74. Skin
75. Hair
76. Nails

Reproductive female system
77. Breasts
78. Uterus
79. Ovaries & tubes
80. Cervix
81. Vagina & clitoris

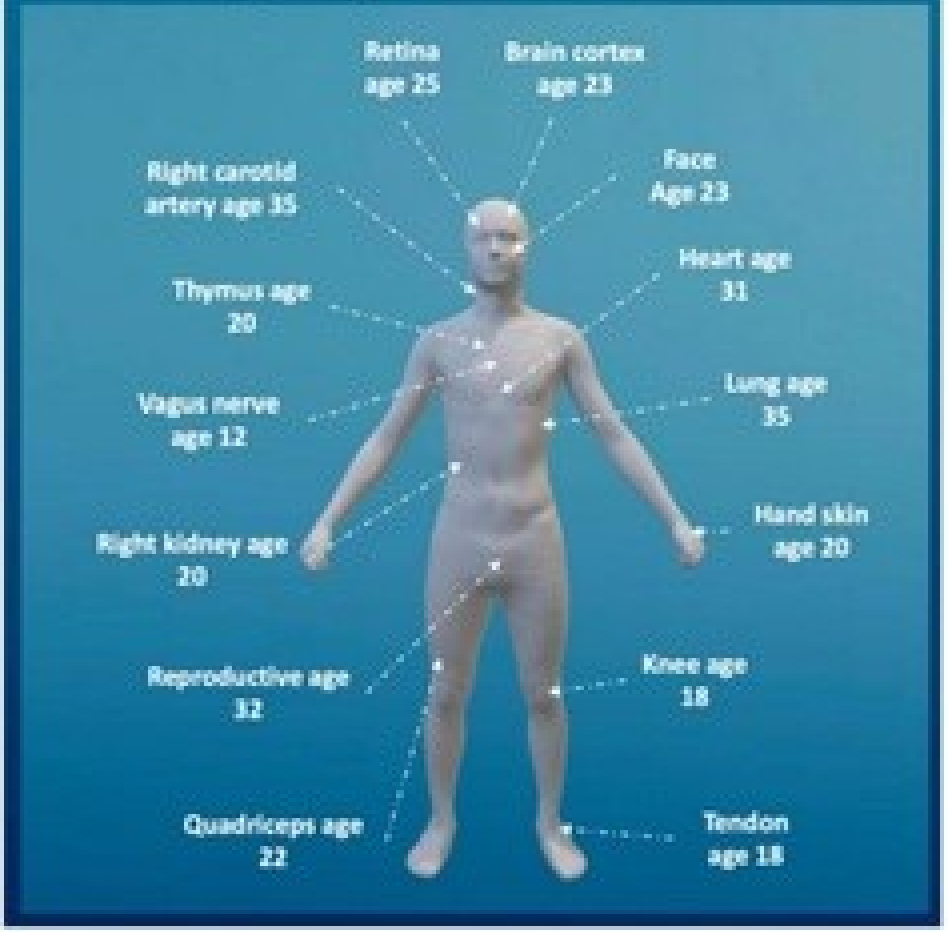
Zolman Ageotypes

Each of 81 organs age at different rates in individuals.

Retina age 25 | Brain cortex age 23

Zolman Ageotypes

Each of 81 organs age at different rates in individuals.



HOW DOES LEVEL 1 DIFFER FROM STANDARD CARE?

WHO: EXERCISE (WHO REF)

WHO 2021 recommends for adults 18+ and 65+
>5 hrs moderate or 2.5 hrs vigorous/week or a blend
3+ days whole body strength training
3+ days whole body strength & balance training if age 65+

VERY SIMILAR, BUT HEART RATE GUIDED METS

AND LIMITING TO 2.5 HRS VIGOROUS

E

D

WHO: DIET, BMI & CRON (WHO REF)

18.5-24.9 BMI for non-Asians, BMI 18.5-22.5 for Asians (+good WHR)
Free sugars <5%, fat <30%, sat fat <10%, trans fat <1%
Fruits+veg >400g excluding potatoes, sweet potato, cassava and other starchy roots
Salt <2g sodium, eat nuts, legumes, whole grains
Calories equal to total energy expenditure

LEVEL 1 DIET IS SIMILAR, MAJOR DIFFERENCE IS CRON

WHO: ALCOHOL

14 units/week limit for both genders, spaced out over at least 3 days
No binge drinking
A binge drink is 8 units for men or 6 units for women in a single session

ESSENTIALLY THE SAME, WITH FOCUS ON ALCOHOL CESSATION

A

S

WHO: SMOKING

Completely stop

IDENTICAL

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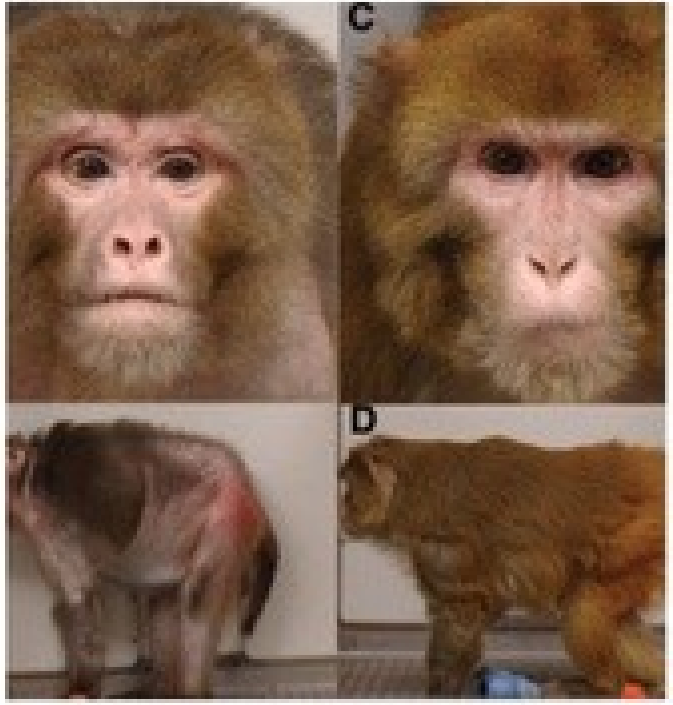
CRON: MONKEY EVIDENCE

- **CRON EXTENDS HEALTHSPAN & LIFESPAN IN MALE RHESUS MONKEYS THAT ARE SIMILAR TO HIGH LEVEL 1 COMPLIANCE**
  - +17% increase in median lifespan, +24% increase in 90th percentile lifespan
  - Median lifespan increased from 29 to 34 (5 years), 90th percentile from 33 to 41 (8 years)
  - 417x more likely to reach rhesus monkey age 40 (equivalent to centenarian status in humans)
  - Longest lived rhesus monkeys ever on record
  - Monkeys started at age 45 human equivalent with no change in older age
  - These monkeys didn't smoke, were alcohol abstainers, had optimal BMI and had a low advanced glycation end product, seemingly high AHEI scoring diet (study) (background info).
  - Full exercise not allowed, but exercise wouldn't cause these results.



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GENE THERAPIES SHOWN TO REJUVENATE IN MICE NOW USED IN HUMANS

1. HTERT (MAKE TELOMERES LONGER) +40% LIFESPAN IN MICE
2. FOLLISTATIN (STOP MUSCLE BREAKING DOWN) +30% LIFESPAN IN MICE
3. KLOTHO (IQ HORMONE DECLINES WITH AGE) +20% LIFESPAN IN MICE

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TRIPLE HELIX
www.triplehelixscience.com

LONGEVITY LEVEL 2

TRIPLEHELIX FOLLISTATIN AAV9 GENE THERAPY

Single IV injection TripleHelix AAV9 Follistatin Gene Therapy

+ 7.2 kg muscle (+15.8 lbs) (+10% muscle as % of total body mass) (DEXA)

Note: my frame is small for my organs (?) hence the bowed appearance (?pectus carinatum)

Body weights
Before: 78.5kg (173 lbs)
After: 78.5 kg (173 lbs)

No TRT/other confounds

VO2 MAX (EXERCISE CAPACITY)

EACH ML = 45 DAYS LIFESPAN

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LEVEL 1 CASE STUDY: DAVID FISHER, 61 YEARS OLD

Started CRON at age 30

1 INTERVENTIONS

1700 kCal/day no breakfast, 700 lunch, 1000 evening meal) (source, 2019)
Wears continuous glucose monitor
Donates blood to keep TSAT at low end
All Longevity Level 1 therapies

2 BIOMARKERS

Coronary Calcium CT Scan score of zero
Balance test top 2%
Visceral abdo fat lowest 2% (MRI AMRA scan)
Reduced calories over 5 years slowly
66 kg height 1. 78 m
BMI 21 (source, 2004 and 2009)

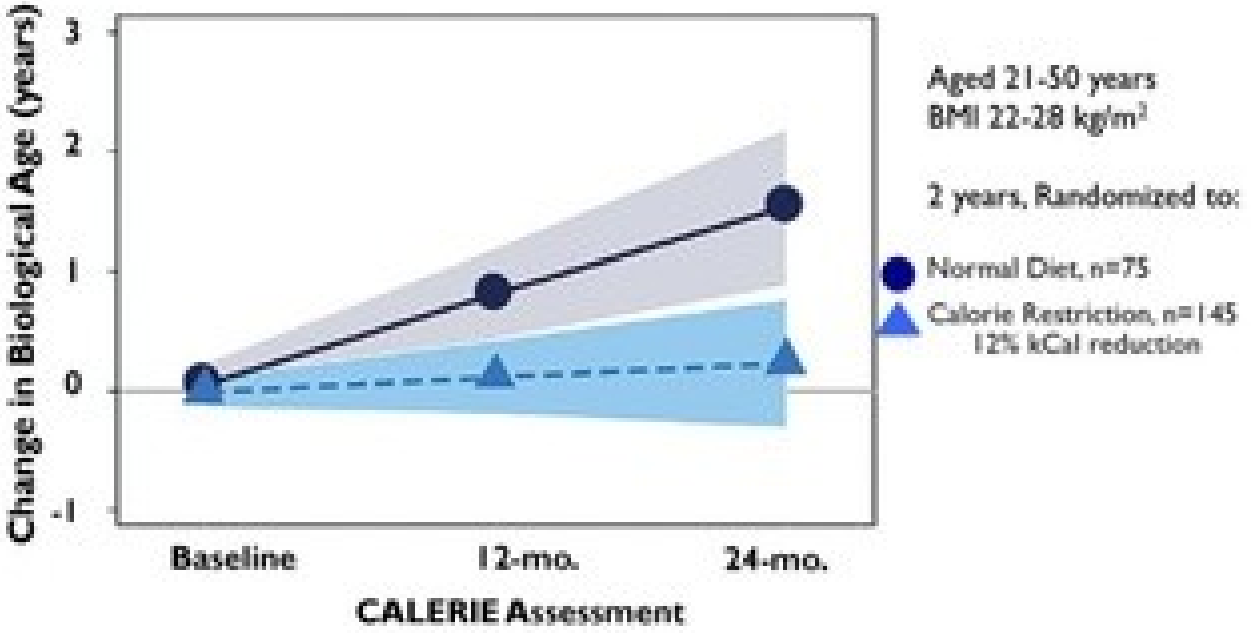
DAVE FISHER IN 2019, AGE 61

SOURCE OF IMAGE: WWW.CRVITALITY.COM/2019/04/DAVID-FISHER-INTERVIEW-CALORIE-RESTRICTION-2019/

2017

CRON SLOWS AGING IN BMI 22-28

Biological aging composite attenuated by 2 years caloric restriction in healthy adults



Belsky DW, et al. J Gerontol A Biol Sci (2017): doi:10.1093/gerona/glx096

LEVEL 1

PERFECT ORAL HEALTH

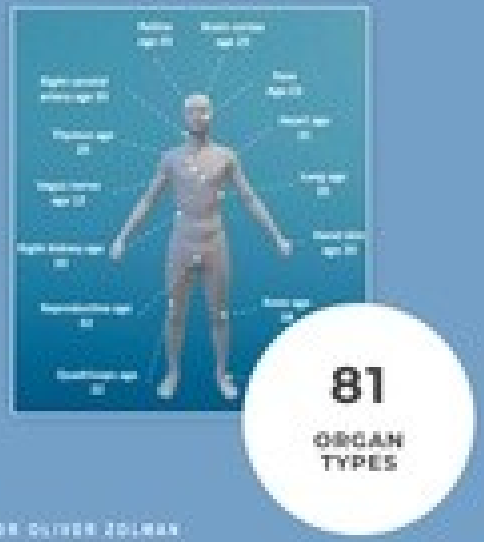
INTRODUCTORY CONCEPTS

LEVEL 1

MENTAL HEALTH

INTRODUCTORY CONCEPTS

LONGEVITY LEVEL 3: REJUVENATE 81 ORGAN TYPES



1. **MEASURE AGE OF EACH ORGAN ACCURATELY**
   81 organ types across both genders
   (And measure for all ICD-11 ageing diseases)

2. **ADDRESS 8 FUNDAMENTAL CAUSES OF AGEING**
   Too few cells, too many cells, damaged cells, waste in cells, waste outsside cells, damaged scaffold, gene expression changes, bad comms between cells

3. **USE ON LABEL, OFF-LABEL AND CLINICAL RESEARCH BASED THERAPIES TO REJUVENATE**
   Can include statins to mounjaro to gene therapy, nanotechnology & organ replacement

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ZOLMAN LONGEVITY LEVEL 1-2-3

LEVEL

1

EVIDENCE BASED ESSENTIALS

Extend Lifespan to avg 94 Yrs (???*)

LEVEL

2

OPTIMISE NON-AGE RELATED FACTORS

Goal: cure every *non-aging* condition

LEVEL

3

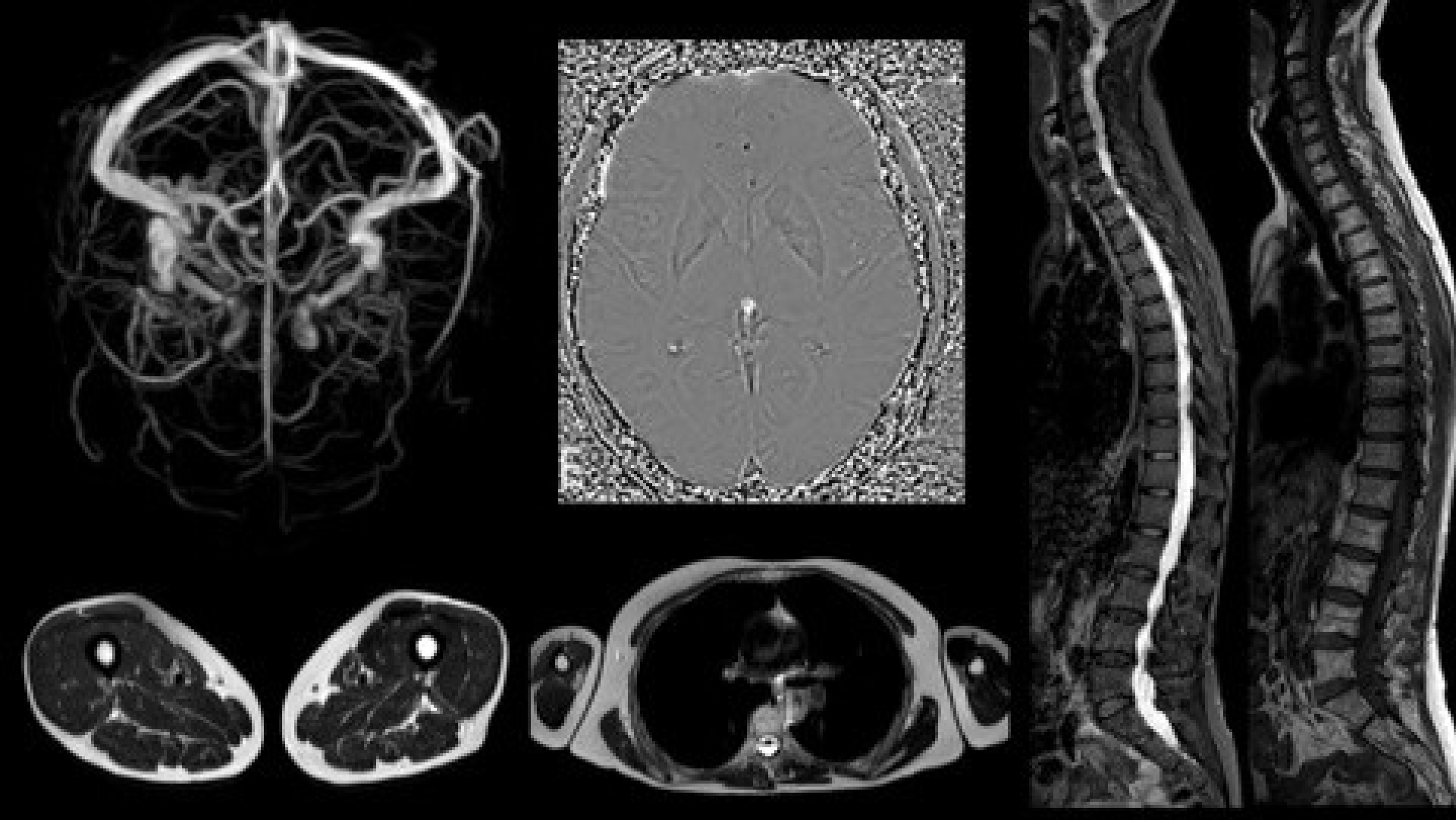
REJUVENATE 81 ORGAN TYPES

Theoretically Unlimited Extension of Healthspan

*NOT RCT PROVEN

STATINS

The Ultimate Gene Therapy: The Terminator
FST: 15% more Muscle Mass
hTERT and Klotho: Brain Aging Reversal
hTERT: Cellular Ageing Reversal
FGF21/PGC-1α: Perfect Metabolic Health
Klotho: Optimal Kidney, Brain, and Vascular Health
FOXO3: Prevents Future Health Decline



100 PEOPLE AGE 75-85 TO UNDERGO FULL EXPERIMENTAL LEVEL 123 PROTOCOL

LEVEL 1 + 2 CASE STUDY: MICHAEL RAE, 49 YEARS OLD

Started CRON age 30

- **INTERVENTIONS**

  ~1900 Cal/day CRON diet 19 years
  3 isocaloric evenly spaced meals/day
  Daily cronometer tracking
  Tailored supplement regimen
  Likely all Level 1 & 2 therapies

- **BIOMARKERS**

  Height 1.82
  Weight 52 kg
  BMI 15.7

SOURCE

LEVEL 1

SLEEP

LEVEL 1

GUIDELINE RECOMMENDED SCREENING TESTS